

HARSH KUMAR

+91-8310321521 ◇ Bangalore, Karnataka

kumarharsh3014@gmail.com ◇ LinkedIn ◇ GitHub

EDUCATION

BTech in Computer Science (HONS) specializing in AI and ML, Manipal University Jaipur

2022-2026

Relevant Coursework: DSA, OOPs, RDBMS, Artificial Intelligence

Soft Computing, Data Science, ML, NLP, Computer Vision, RL

CGPA: 8.87/10

SKILLS

Technical Skills

Python, MySQL, Large Language Models (LLMs), Neural Networks, NLP, Retrieval-Augmented Generation (RAG)

Frameworks & Libraries

LangChain, HuggingFace Transformers, Scikit-Learn, TensorFlow, PyTorch, OpenCV

Developer Skills

AWS tools, Git, API integrations, CI/CD (theoretical), Azure DevOps (theoretical)

EXPERIENCE

Quest Global - AI Intern

May 2025 - July 2025

- Built a context-enriched **Retrieval-Augmented Generation (RAG)** pipeline using **multiple LLMs**(LLaMA, Mistral) for intelligent test case generation, achieving **80–90%** accuracy vs. manual cases, plus identifying critical edge cases, reducing manual effort by over **70%**.
- Embedded UI layout metadata into **LLM reasoning**, enabling richer **semantic coverage** and precision in generated test cases.
- Built an automated test execution **AI agent** integrating **AWS Strands agents SDK, AWS Bedrock, Claude, MCP, and Playwright**, enabling fully autonomous test validation.
- Delivered scalable, **AI-driven QA automation** that improved efficiency, overall cutting manual workload by **60%**, and accelerating testing cycles by **30%**.

Action Tour Guide - Python Developer Intern

June 2024 - July 2024

- Developed and deployed **Python automation scripts** to efficiently extract, process, and organize data from customer emails, reducing manual effort by **40%** and ensuring accuracy in data handling.
- Designed and implemented automated **Python workflows** to schedule and send follow-up emails post-tour completion, streamlining the process and boosting **feedback response rates** by **25%** while ensuring timely customer engagement.

PUBLICATIONS

Chronocept: Instilling a Sense of Time in Machines 

EACL SRW 2026

- Co-authored **Chronocept**, the first benchmark to model **temporal validity** as a continuous probability distribution, using skew-normal curves over a logarithmic time axis to capture **emergence, peak relevance, and asymmetric decay** of factual knowledge.
- Designed a structured temporal annotation framework grounded in **MATRES** decomposing facts across **8 semantic temporal axes**, and curating two benchmarks(Atomic facts and Multi-Sentence passages) with **1700+** samples and high inter-annotator agreement(**84–89%** IAA).
- Developed and benchmarked geometry-aware regression SOTA models (**RoBERTa, DeBERTa-v3, MT-DNN, SBERT variants**) using a **parameter-space Gaussian NLL** objective to jointly predict **location, scale, and skewness**, revealing trade-offs between accuracy, calibration, and temporal generalization.

PROJECTS

Lexify - Indian Law Legal Assistant — (Python, Groq API, ChromaDB, Streamlit)

- Built an AI legal research tool using **LLaMA3-8B (Groq)** to process **500+** IPC, Constitution, and Labour Act sections with **ChromaDB + HuggingFace embeddings**.
- Achieved a **2–4s response times** via model optimization, document chunk caching, and **LangChain-based** multi-step query pipelines.
- Created a **Streamlit UI** with session persistence and styled query history for reviewing **30+** past interactions.

Stock Market Prediction — (LSTM-based, Time Series Models)

- Developed and trained a **Long Short-Term Memory (LSTM)** model for **NIFTY50 stock price prediction**, utilizing historical market data, technical indicators (e.g., moving averages, Bollinger Bands), and time series analysis techniques.
- Attained a score of **R² of 0.96**, significantly improving stock price forecasting accuracy and providing actionable insights for investment decisions.

MedVision-LLM: AI Medical Diagnosis System — (PyTorch, Ensemble Deep Learning, Streamlit, GPT-4)

- Engineered ensemble deep learning system (**EfficientNet-B3 + ResNet50 + DenseNet121**) for 4-class chest X-ray classification, achieving **96.22% accuracy**, and across **10,000+** medical images
- Implemented **Grad-CAM explainability** with backward hooks on final convolutional layers and integrated **GPT-4 API** to auto-generate structured radiology reports, reducing interpretation time from minutes to **<2 seconds** per case.
- Optimized training with **focal loss, mixed-precision (FP16)**, and **cosine annealing (40% reduction)**; deployed **Streamlit** application achieving **<2s latency**.

ACHIEVEMENTS AND CERTIFICATIONS

- 2 x Dean's List for Academic Excellence
- Machine Learning Specialization by DeepLearning.AI, Stanford by Andrew Ng.
- Finalist for National SAP Hackathon.

- Data Structures Certification from UC San Diego on Coursera
- CCNAv7 Cisco Certification on Introduction to Packet Tracer and Enterprise Networking, Security and Automation and Switching, Routing and Wireless Essentials
- Swayam NPTEL Course Certification for Machine Learning

LEADERSHIP RESPONSIBILITIES

Randomize - Treasurer

2023-2025

- As **Treasurer in 2024**, I managed club finances and spearheaded the organization of our club's Annual **Hackathon**, and also worked on multiple projects while mentoring the juniors in the club on AI/ML.
- As **Head of IT and Operations**, led core committee operations, and event flow for major tech events(**Hackathons, Hello World, etc.**), resulting in seamless execution of all events.
- Also delivered a high impact talk on **Data Structures and Algorithms(DSA)** and the domain of **AI, covering Machine Learning, Deep Learning, Reinforcement Learning, Gen AI, and LLMs**.

Turing Sapiens - Head of Projects

2023-2024

- As **Head of Projects**, worked on building predictive analysis models and Deep Learning classification models for the club.
- Helped in organizing the annual fest - **Alphasia**, co-leading in organizing the hackthon for several team from college.